

Supplementary Information

Beyond Per Capita: Superlinear Scaling and Industrial Decoupling in Urban Water Metabolism

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1 Water use CR2WU

The database employs two methodologies to estimate water use: one for land water use (LWU), i.e. sectors related to land use, land-use change, and forestry (LULUCF), and the other for non-land water use (NLWU) sectors. LWU is estimated as water budget-based approach and for NLWU, the flow used by sector (U_s) is derived by weighting the activity-specific consumption rate and corresponding forcing data (Equation S1). Sources of information include: the DGA, SISS, INE, and Central Bank (Boisier et al., 2025).

$$U_s(x, yr) = \sum_s K_s(x) \cdot F_s(x, yr) \quad (S1)$$

x commune, yr year, s sector

U_s Volumetric flux use (e.g., L/year)

K_s Sector-specific consumption rate (e.g., L/person/year)

F_s Forcing (e.g., Population)

For example, Drinking Water (DW) is the sum of the following subsectors of urban use: domestic (residential use), green areas, commercial, and industrial (water use for commerce and industries within urban

areas, which are not included in the Industrial sector), as well as the domestic rural. Each of these sub-sectors has specific consumption rates per subsector and per commune, drawn either from empirical data (e.g., drinking water supply billing records) or estimates (e.g., domestic rural consumption by northern, central, and southern zones). Figure S1 shows the relationship between these consumption rates that constitute the DW sector and population.

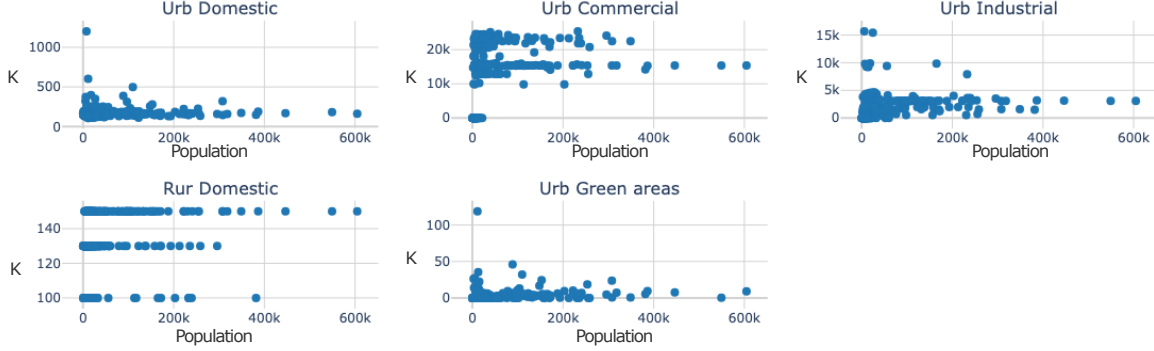


Fig. S1 Drinking Water subsector-specific consumption rates by population for all Chilean communes for the year 2017

2 Predictor model variables

Preliminary explanatory variables for the multiple and scaling models of water use at the commune level are divided into two groups: (i) demographic and environmental data (Population, Density, Precipitation, Temperature, Latitude), and (ii) socioeconomic data (ISMT, Firms, Revenue, Employees, Revenue/firm, Workers).

2.1 Demographic and environmental data

In the Figure S2, we illustrate the distribution of demographic and environmental data, including the log-transformed values for the 343 continental communes of Chile.

Annual population and population density at the level of communes were derived from logistic-based projections provided by the National Statistics Institute (INE, 2021). Latitude was included as a geographical variable to capture spatial patterns across Chile’s extensive latitudinal gradient. Environmental characterization included two meteorological variables: annual mean temperature (Temp) and cumulative annual precipitation (Precip), both sourced from the CR2MET climate dataset. Since station-based meteorological data represent point-scale measurements, we estimated communal-level values from this gridded product. CR2MET provides climate variables on a 0.05° regular grid across Chile, integrating statistically calibrated ECMWF ERA5 reanalysis data, topographic parameters, and MODIS land surface temperature (Boisier, 2023). Communal zonal statistics were derived from these monthly gridded rasters; specifically, monthly values were area-weighted and averaged across raster cells within each administrative boundary. These were then temporally aggregated into annual totals for precipitation and yearly means for temperature. All geospatial processing and raster extractions were conducted using the terra package in R (Hijmans, 2025).

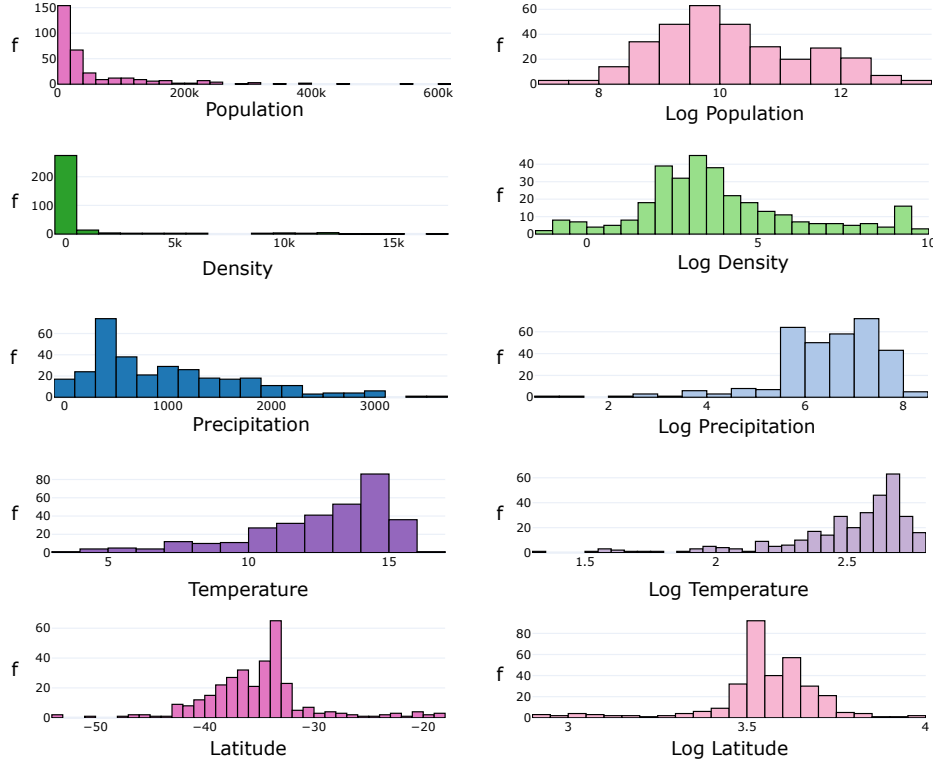


Fig. S2 Demographic and environmental data distributions for the year 2017. Histograms depict raw and log-transformed communal-level values across the 343 continental communes.

2.2 Socioeconomic data

Indicators representing local socioeconomic characteristics are not officially compiled for communes. For instance, while Gross Domestic Product (GDP) serves as a primary metric of economic activity at national and regional scales, commune-level GDP estimates are not officially computed in Chile. Consequently, we employed a suite of alternative socioeconomic proxies: the Territorial Sociomaterial Index (ISMT), the number of companies, and the annual sales of companies, the number of employees in companies (tax data from the Chilean Internal Revenue Service, SII), and the number of workers reported in the National Socioeconomic Survey (CASEN) (CASEN, 2017; SII, 2024)

The ISMT is a composite index used as a socioeconomic proxy across geographical and demographic scales in Chile. Developed by the Observatorio de Ciudades UC (OCUC), the index integrates four key variables derived from census data: education, household overcrowding, urban proximity, and housing materiality. These four variables are derived from national survey data at the tract level (which constitutes the primary unit of a census district). Although the ISMT cannot be classified as a socioeconomic indicator per se (according to the ISMT’s methodology), it represents a useful index because it integrates, in a single indicator, socioeconomic and material conditions of the territory at a fine scale (OCUC, 2019).

To gather communal socioeconomic data, we utilized the 2017 ISMT database. To address the high levels of intra-communal socioeconomic segregation (Fig. S3), we calculated the aggregate index for commune i as a population-weighted mean of its census tracts j . This approach ensures that the resulting value (Eq. S2) accurately reflects the internal distribution of the population:

$$ISMT_i = \frac{\sum_{j \in Z_i}^{n_i} (ISMT_{ij} \cdot P_{ij})}{\sum_{j \in Z_i}^{n_i} P_{ij}} \quad (S2)$$

where:

- i denotes the commune and j the census tract;
- Z_i represents the set of all census tracts within commune i ;
- P_{ij} is the population of census tract j within commune i .

To approximate industrial productivity, enterprise development data were sourced from 2017 firm tax filings. Three primary variables were extracted: total number of firms (Firms), annual revenue in USD (Revenue), and total number of employees (Employees). We further derived the revenue per firm (Rev/firm) as a proxy for communal productivity. To address potential spatial bias—specifically, that tax declarations (RUT) are typically registered to a firm’s head office rather than its operational production sites—we incorporated the number of individuals physically employed within the commune from the CASEN survey (Workers). This serves as a complementary proxy for local economic activity. Figure S4 illustrates the frequency distributions and log-transformed values for these variables across the 343 continental communes studied.

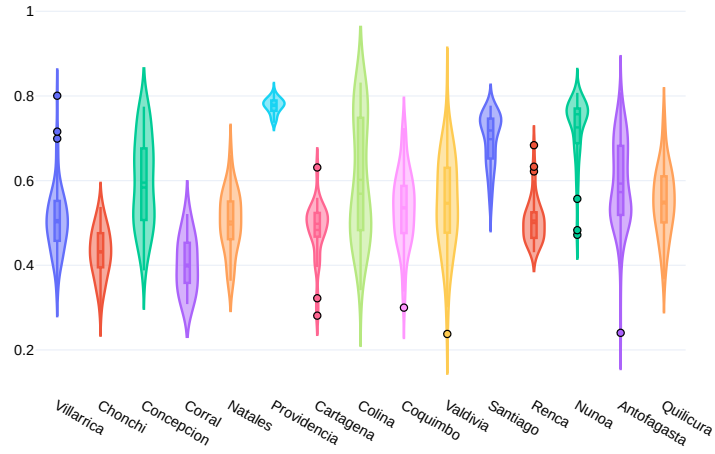


Fig. S3 ISMT distribution for a sample of communes. For example, Colina exhibits considerable variability, suggesting significant socioeconomic segregation. Here, some census areas have a high socio-material index (close to 0.9) while others show a low index (below 0.3).

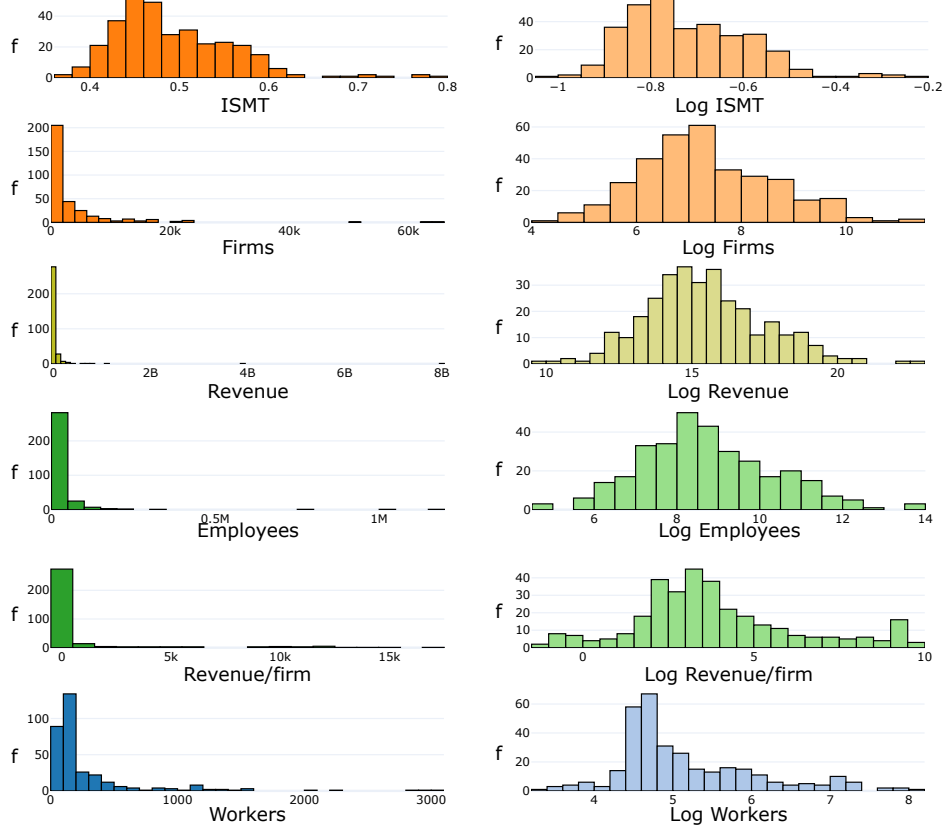


Fig. S4 Socioeconomic data distributions for the year 2017. Histograms depict raw and log-transformed communal-level values across the 343 continental communes.

2.3 Predictor relationships

The correlation analysis revealed a strong link between socioeconomic variables and population, and a high correlation between latitude and precipitation, which aligns with expectations (see spearman correlation heatmap in Fig. 2(B)). This outcome is expected, as more populous communes are expected to have a higher concentration of companies and workers, and the southernmost communes typically experience greater rainfall. Based on the pairwise correlations, we excluded only latitude, since our goal was to retain all potential socioeconomic predictors.

From the original 11 variables, we assessed multicollinearity among 10 (excluding latitude) with Variance Inflation Factor (VIF). When VIF is higher than 10, there is significant multicollinearity that needs to be corrected (See Table S1).

As expected, Firms, Revenue, and Revenue per firm showed high VIF values, as the latter is a function of the first two. We thus kept only Revenue per firm and recalculated VIF for the remaining eight variables. Two variables had $VIF > 10$: Employees (14) and Population (13). After removing Employees, all seven remaining variables had $VIF < 10$.

The variable Workers was discarded because data was unavailable for 20 communes, resulting in a final set of six potential predictor variables: Population, Density, ISMT, Rev/firm, Precipitation, and Temperature.

3 Model results

The scaling model is solved as a linear regression of the log-log model of three types of water use (Drinking Water, Industrial, and the sum of both, referred to as Non-Land Water Use, NLWU) against population.

Table S1 Variance Inflation Factor (VIF)
values of explanatory variables

Variable	VIF 1	VIF 2	VIF 3
Firms	∞	—	—
Revenue	∞	—	—
Rev/firm	∞	2.89	1.79
Population	21.57	13.74	8.41
Employees	19.64	14.09	—
Workers	4.98	4.96	4.86
Density	4.62	4.39	4.38
ISMT	3.39	2.99	2.78
Temperature	2.26	2.89	2.20
Precipitation	1.67	1.61	1.58

This table includes the scaling coefficient (β) with respect to population estimated using Ordinary Least Squares (OLS) and quantile regression at the 10th and 90th percentiles. As many communes have no industrial water use, three treatments were evaluated for this variable: Industrial (non-zero) omits zero observations and applies the natural logarithm. Industrial+1 and Industrial+c keep all communes and use the shifted logarithmic transformations $\log(x+1)$ and $\log(x+c)$, where $c = 1e-7$ is a small constant set one order of magnitude below the smallest positive industrial water use value. The outcomes of the scaling model are reported in Table S2.

Table S2 Linear regression results for scaling model, with population as the explanatory variable and water use as response variable for three types of use: Drinking Water (DW), Industrial, and Non Land Water Use (NLWU).

Variable	N	Model	R^2	F	β	SE	$p(\beta=0)$	$p(\beta=1)$	CI 95%
NLWU	343	OLS	0.58	377.68	0.87	0.04	<0.001	0.005	[0.78, 0.96]
		Q10				1.05	0.01	<0.001	[1.02, 1.07]
		Q90				0.62	0.13	<0.001	[0.37, 0.87]
DW	343	OLS	0.97	5979.47	1.08	0.01	<0.001	<0.001	[1.05, 1.10]
		Q10				1.09	0.01	<0.001	[1.08, 1.11]
		Q90				1.07	0.03	<0.001	[1.01, 1.13]
Industrial (non-zero)	341	OLS	0.02	4.1	0.23	0.12	0.044	<0.001	[0.01, 0.46]
		Q10				-0.14	0.30	0.640	[-0.73, 0.45]
		Q90				0.41	0.05	<0.001	[0.30, 0.51]
Industrial+c	343	OLS	0.01	1.3	0.15	0.13	0.255	<0.001	[-0.11, 0.40]
		Q10				-0.18	0.34	0.589	[-0.85, 0.48]
		Q90				0.41	0.05	<0.001	[0.30, 0.51]
Industrial+1	343	OLS	0.05	23.06	0.09	0.02	<0.001	<0.001	[0.05, 0.13]
		Q10				0.00	0.02	0.947	[-0.03, 0.03]
		Q90				0.28	0.03	<0.001	[0.22, 0.35]

4 Scaling interpretations

The multiple model for Non-Land Water Use (NLWU)—comprising aggregated residential and industrial consumption—is formulated as a Cobb-Douglas-type production function. In this framework, NLWU is modeled as a product of specific explanatory factors, each representing an input into the commune's total water demand. The model identified five primary independent variables:

$$NLWU = Pop^{\beta_1} \cdot Dens^{\beta_2} \cdot Precip^{\beta_3} \cdot Rev/firm^{\beta_4} \cdot ISMT^{\beta_5}$$

The linearized equation is:

$$\log(NLWU) = \beta_1 \log(Pop) + \beta_2 \log(Dens) + \beta_3 \log(Precip) \\ + \beta_4 \log(Rev/firm) + \beta_5 \log(ISMT) + \epsilon$$

We will analyze three predictor variables representing three patterns: superlinear ($\beta > 1$), sublinear ($\beta < 1$), and negative scaling ($\beta < 0$).

- Population, $\beta > 1$

What would be the expected consumption of a commune with **twice** the population of another, assuming all other explanatory variables remain equal (*ceteris paribus*)?

$$\log(NLWU) = 1.14 \cdot \log(Pop) + \epsilon'$$

The difference between water use of both communes will be:

$$\log(NLWU_2) - \log(NLWU_1) = 1.14 \log(Pop_2) - 1.14 \log(Pop_1)$$

Considering that $Pop_2 = 2 \cdot Pop_1$

$$\begin{aligned} \log(NLWU_2) - \log(NLWU_1) &= 1.14 \log(2Pop_1) - 1.14 \log(Pop_1) \\ \log(NLWU_2) - \log(NLWU_1) &= 1.14 \log(2) + 1.14 \log(Pop_1) - 1.14 \log(Pop_1) \\ \log(NLWU_2) - \log(NLWU_1) &= 1.14 \log(2) \\ \log(NLWU_2/NLWU_1) &= 0.79 \\ NLWU_2/NLWU_1 &= e^{0.79} \\ NLWU_2/NLWU_1 &= 2.2 \\ NLWU_2 &= 2.2 \cdot NLWU_1 \end{aligned}$$

Then, a commune that doubles the population of another will consume more than double the amount of the first one (an additional 20%).

- ISMT, $\beta < 1$

What would be the expected consumption of a commune with **twice** the 'wealth' (proxied with ISMT) of another, assuming all other explanatory variables remain equal?

$$\log(NLWU) = 0.35 \cdot \log(ISMT) + \epsilon'$$

Considering that $ISMT_2 = 2 \cdot ISMT_1$, the difference between water use of both communes will be:

$$\begin{aligned} \log(NLWU_2) - \log(NLWU_1) &= 0.35 \log(2 \cdot ISMT_1) - 0.35 \log(ISMT_1) \\ \log(NLWU_2) - \log(NLWU_1) &= 0.35 \log(2) + 0.35 \log(ISMT_1) - 0.35 \log(ISMT_1) \\ \log(NLWU_2) - \log(NLWU_1) &= 0.35 \log(2) \\ \log(NLWU_2/NLWU_1) &= 0.24 \\ NLWU_2/NLWU_1 &= e^{0.24} \\ NLWU_2 &= 1.27 \cdot NLWU_1 \end{aligned}$$

Consequently, a commune that is twice as 'affluent' (according to the ISMT) as another consumes less than twice as much water (only 27% more). However, this does not imply that wealthier communes consume less domestic water per capita.